

Final Lab Exam (CPCS-204)

Fall 2024

Total Time: 100 Minutes

Fall 2024

Max. Marks: 15

Name: _____ **ID:** _____ **Section:** _____

Note: Read the following instructions very carefully!

1. From **Blackboard**, download **Final Lab Exam**.
 2. Save the .rar or .zip folder on Desktop or at your desired location.
 3. Unzip this folder by right clicking on this folder.
 4. Open **NetBeans** and select the option **File Open Project...** from the main menu.
 5. Open all classes of your lab exam and **write your ID, Name** etc on top of every class and start writing the code in the classes given below.
 6. Keep on saving your files after some time.
 7. When you finish your exam, go to Blackboard and upload your solution after zipping the folder which contains your solution.
 8. **Only students are responsible for uploading the right solution folder on Blackboard.**
- No invigilator is responsible to do so.**

You can add or delete any parameter in any class according to your requirements.

LinkedList class	Total [5 marks]
Complete the functionality of the following method: passed() Method • Suppose we have a single link list containing the marks of students for a course. • passed() method generates a NEW list containing ONLY the students who PASSED the course. Assume that student will be considered passing if he got 60 or more marks. • Upon completion, the original list should be the same. The new list should have only those students who passed the course. [5 Marks]	

intBST class	Total [5 marks]
Complete the functionality of the following method: findShortestLL() Method • Write code to visit each BST node and count the values of its respective Linked-List and then return the BST node “value” whose Linked-List is the shortest. [5 Marks]	

BSTnode class	Total [2 marks]
Complete the functionality of the following method: getCountInRange() • Given a Binary Search Tree (BST) and a range [u, v]. • getCountInRange() method counts the number of nodes in the BST that lie in the given range. [3 Marks]	

FinalLabExam Fall2024 class	Total [3 marks]
P1. In choice 3 in the next sample output, write a method in intBST class to get the “value” from the user (save into variable “LLvalue”) and then to delete the Linked-List of the BSTnode whose value is “value” and delete this BSTnode. [2 Marks]	
P2. In choice 5 in the next sample output, write statements to get the number of nodes in BST that lie in a given range [a, b] and print it. [1 Mark]	

Sample output:

run:

```
|-----|
|----- Binary Search Tree & Linked List Menu -----|
|-----|
| 1. Insert a new BST node into the tree |
| 2. Insert a new Linked-List node (value) into a specific node of the BST |
| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 1
Enter value to insert into BST: 4
.

|-----|
|----- Binary Search Tree & Linked List Menu -----|
|-----|
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| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 1
Enter value to insert into BST: 6
```

```

|-----|
|----- Binary Search Tree & Linked List Menu -----|
|-----|
| 1. Insert a new BST node into the tree |
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 1
Enter value to insert into BST: 8

|-----|
|----- Binary Search Tree & Linked List Menu -----|
|-----|
| 1. Insert a new BST node into the tree |
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| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 2
Enter BST node value to insert into: 4
Enter value to insert into Linked-List: Enter the number of values to insert into the LinkedList: 2
Enter value 1: 5
Enter value 2: 6
Values inserted into the LinkedList successfully.

```

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|-----|
|----- Binary Search Tree & Linked List Menu -----|
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 2
Enter BST node value to insert into: 6
Enter value to insert into Linked-List: Enter the number of values to insert into the LinkedList: 5
Enter value 1: 2
Enter value 2: 3
Enter value 3: 0
Enter value 4: 4
Enter value 5: 5
Values inserted into the LinkedList successfully.

```

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|-----|
|----- Binary Search Tree & Linked List Menu -----|
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 2
Enter BST node value to insert into: 8
Enter value to insert into Linked-List: Enter the number of values to insert into the LinkedList: 3
Enter value 1: 5
Enter value 2: 8
Enter value 3: 10
Values inserted into the LinkedList successfully.

```

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|-----|
|----- Binary Search Tree & Linked List Menu -----|
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| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|

```

Enter your choice: 4

BST node with the shortest linked list has value: 4

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|-----|
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| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|

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Enter your choice: 5

Enter lower value range: ...2

Enter higher value range: ...7

The counts of node in range [2,7] = 2

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|-----|
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 6
Here is your tree:
4: 5 6
6: 2 3 0 4 5
8: 5 8 10

```

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|-----|
|----- Binary Search Tree & Linked List Menu -----|
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 3
Enter BST node value to delete it and its linked list: 6
Node with value 6 and its LinkedList have been deleted.

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|-----|
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 6
Here is your tree:
4: 5 6
8: 5 8 10

```

```

|-----|
|----- Binary Search Tree & Linked List Menu -----|
|-----|
| 1. Insert a new BST node into the tree |
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 7
Enter how many student marks you want to put: 10
Enter student 1 Marks : 60
Enter student 2 Marks : 65
Enter student 3 Marks : 50
Enter student 4 Marks : 70
Enter student 5 Marks : 100
Enter student 6 Marks : 89
Enter student 7 Marks : 55
Enter student 8 Marks : 80
Enter student 9 Marks : 69
Enter student 10 Marks : 85

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|-----|
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| 3. Delete a Linked-List node (value) into a specific node of the BST |
| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 8
60 65 70 100 89 80 69 85

```

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|-----|
|----- Binary Search Tree & Linked List Menu -----|
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| 4. Find which BST node has the shortest linked list of values |
| 5. Count the number of nodes in range a & b in the BST |
| 6. Print tree |
|-----|
| 7. Insert a new marks into student Linked list |
| 8. Generates a NEW list containing ONLY the students who PASSED the course |
| 9. Quit |
|-----|
Enter your choice: 9
Exiting...

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BUILD SUCCESS
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Total time: 03:15 min
Finished at: 2024-11-19T00:35:41+03:00
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